

Statistics

Lecture 06

Confidence intervals for unknown model parameters

Parameter estimation reviewed

The concept of Confidence Intervals (CI) and confidence level

Case: CI for μ for $X_i \sim N(\mu, \sigma^2)$, σ known

- model and observations
- estimate for μ and its distribution
- 2-sided CI for μ and interval width, **norminv**
- 1-sided upper (lower bounded) and lower (upper bounded) CI's for μ

Case: CI for μ for $X_i \sim N(\mu, \sigma^2)$, σ unknown

- model, observations, estimate for μ and its distribution
- definition of χ^2 (chi squared) distribution and its properties
- estimate $s^2 = \hat{\sigma}^2$ for the unknown variance and its distribution
- definition of t (Student t) distribution and its properties
- 2-sided CI for μ and interval width, **tinvs**

Case: CI for σ^2 for $X_i \sim N(\mu, \sigma^2)$

- estimate for σ^2 and its distribution
- 2-sided CI for σ^2 , **chi2inv**

Case: CI for p for $X \sim \text{Bin}(n, p)$ using normal distribution approximation

- model and observation
- estimate $\hat{p}$ and its exact distribution
- approximate distribution for $\hat{p}$ for $n \gg 1$ (and $np, n(1-p) \gg 1$)
- 2-sided CI for p , **norminv**
- same approach can be used for CI for λ if $X \sim \text{Poisson}(\lambda)$

Case: CI (prediction interval) for $(n+1)$ 'th (next) observation for $X_i \sim N(\mu, \sigma^2)$

- estimate $\hat{X}_{n+1} = \bar{X}$ and its distribution
- next observation X_{n+1} and its distribution
- 2-sided prediction interval for X_{n+1} , **tinvs**